

The product I imagined

A pre-triaged morning queue that helps investigators spend attention where one check can change the decision

USER AND CORE JOB

Fraud investigator starting a morning queue

Find the referrals that need real attention, confirm the evidence quickly, and close likely false positives safely.

The system prepares the case. The investigator still owns the decision.

THE MORNING QUEUE

WHAT I BUILT

Three triage lanes
Checked case brief and evidence
Recommended first action
Case Q&A, review, notes, decision log
Audit gate before bulk close

WHAT I LEFT OUT

Authentication and roles
Multi-user workflow
Production database
Claims-system integration

These do not change the core AI experience for 50 synthetic claims, so I used the time on triage, evidence, and human control.

8 Priority investigation

13 Needs judgment

29 Likely false positive

No fraud probability is shown because the file has no outcome labels.

Architecture

Deterministic code decides the lane, evidence, and first action; the LLM turns those facts into a readable brief



HOW THE SYSTEM REASONS

1 Transparent index

Signals become queue percentiles; their plain average is a descriptive index.

2 Groups before labels

Two gaps form 30 / 12 / 8 groups; safety rules produce 29 / 13 / 8 lanes.

3 Statistical cross-check

PC1 explains 74.5%; index-PC1 $r = 0.999$. These do not measure fraud accuracy.

HOW THE AI IS KEPT GROUNDED

1 One constrained call

The model writes from a fixed fact sheet; it cannot choose lane, evidence, or action.

2 Checks and fallback

Checks cover numbers, entities, lane wording, and unsafe claims. Repair once; then use the rule brief.

3 Evidence stays deterministic

Frozen 19-case check: model-selected required evidence 62%; deterministic engine 100%.

Scale: batch and cache briefs; freeze references by care type; version calibration; monitor drift, cost, and latency.

Trade-offs and human control

The system prepares a decision; the investigator validates the evidence and owns the outcome

HOW A HUMAN STAYS IN THE LOOP

1 Trust

Every finding shows the raw value, comparison range, and triggered rule.

2 Validate

The investigator marks each finding agree, disagree, or not sure and can ask case-specific questions.

3 Override

A recommendation never takes action. A different decision requires a written reason.

4 Audit

Targeted and random cases must be reviewed before bulk close. The log keeps the model, prompt, text, and fact fingerprint.

TRADE-OFFS AND NEXT TESTS

Accuracy

No labels: no fraud probability or accuracy claim.
Next: collect outcomes and calibrate.

Trust

Checked evidence and a rule fallback.
Next: blinded investigator study.

Cost and latency

Briefs run before the shift and are cached.
Next: test smaller models at load.

Privacy and abuse

Synthetic data; Q&A blocks pay or deny requests.
Next: approved endpoint and access controls.

With more time Run in shadow mode; measure time to decision and audited misses; freeze calibration; monitor drift.